

Game Design Document

Kitty Kultivation

An urban gardening game set in a world where rapid urbanization is left unaddressed and is evergrowing.

Members:

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1 Introduction

1.1 Game Summary

Kitty Kultivation is a Gardening Simulator Video Game designed to show the players the benefits of indoor gardening within an urban setting. The player will be buying seeds, planting, harvesting and either selling or using the harvest as food. Kitty Kultivation also poses the issue of the potential future if rapid urbanization is left unaddressed.

You've worked a stressful life in your company, and have begun wondering how to improve your life much further from where it is today, embark on a gardening journey on the premise of your own home. The place might be a mess and very smoggy but if you keep your heads up and tools ready, you might be able to change all of that.

1.2 Inspiration

Stardew Valley

Stardew Valley is one of the key inspirations as well for the gardening care methods



Rental

Although a completely different genre of game. Kitty Kultivation takes the character design of Rental as inspiration. The small and cute characters in a world design similar to that of old 3d games is what we took inspiration from.



Project M.T.P

Created by Elias Daler, Project M.T.P is a prototype created to show the capabilities of Vulkan in creating games. Kitty Kultivation took heavy inspiration from Project M.T.P when it came to 3D Designs and world design.

1.3 Platform

The game is played in PC windows and will be available on Itch.io and if possible financially, Steam.



1.4 Development Software

- Unity
- Blender
- Any audio software
- Medibang

1.5 Genre

Singleplayer, Simulation-Farming

1.6 Game Target Audience

The game is suitable for Everyone according to ESRB ratings

2 Concept

2.1 Gameplay Overview

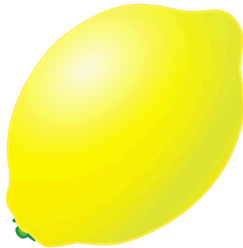
The gameplay is simple, it is generally a farming game with limited space, so space management is one of its mechanics, another part of the game is the

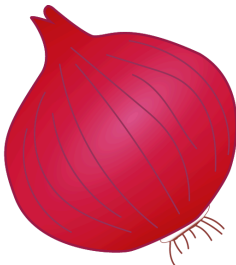
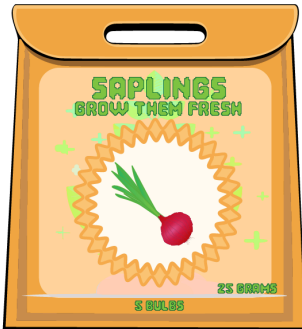
company minigame, where you must do a minigame to earn money and buy items that help improve your indoor garden/farm.

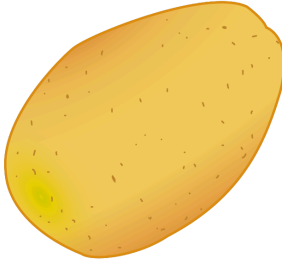
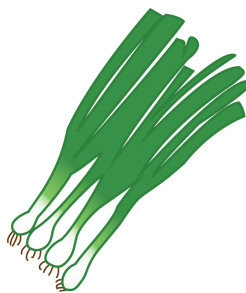
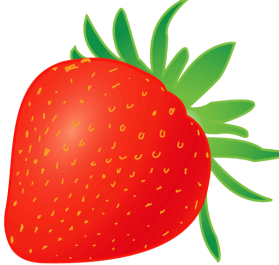
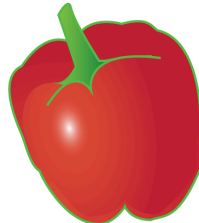
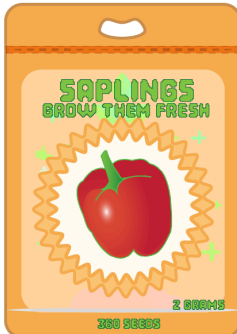
2.2 Primary Mechanics

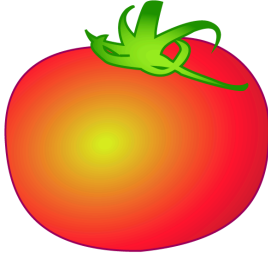
Gardening is the primary game mechanic of this game, however we thought that if we just did farming, our game wouldn't be any different than any other farming game, so we set our setting of our game in an urban area, an apartment building with limited space, you have to plan which plant to farm and where to organize your garden, an open window with a small greenhouse, or a small farm plot next to the window. The player will be able to take care of the garden by weeding out unwanted weeds and watering them. When the harvest is done the player will be able to sell the harvested plants or keep them for self sustenance, as the game contains both a hunger and energy system.

LIST OF PLANTS THAT CAN BE FOUND IN THE GAME

Harvested	Seed Picture	Attributes										
Garlic 	Garlic Seed 	<table><tr><td>Seed Price</td><td>20</td></tr><tr><td>Selling Price</td><td>85</td></tr><tr><td>Days to Grow</td><td>45</td></tr><tr><td>Regrowable</td><td>No</td></tr><tr><td>Days to Re-Grow</td><td>0</td></tr></table>	Seed Price	20	Selling Price	85	Days to Grow	45	Regrowable	No	Days to Re-Grow	0
Seed Price	20											
Selling Price	85											
Days to Grow	45											
Regrowable	No											
Days to Re-Grow	0											
Lemon 	Lemon Seed 	<table><tr><td>Seed Price</td><td>100</td></tr><tr><td>Selling Price</td><td>150</td></tr><tr><td>Days to Grow</td><td>20</td></tr><tr><td>Regrowable</td><td>Yes</td></tr><tr><td>Days to Re-Grow</td><td>13</td></tr></table>	Seed Price	100	Selling Price	150	Days to Grow	20	Regrowable	Yes	Days to Re-Grow	13
Seed Price	100											
Selling Price	150											
Days to Grow	20											
Regrowable	Yes											
Days to Re-Grow	13											

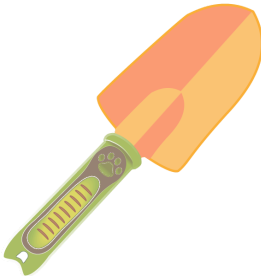
Lettuce 	Lettuce Seed 	<table><tr><td>Seed Price</td><td>50</td></tr><tr><td>Selling Price</td><td>50</td></tr><tr><td>Days to Grow</td><td>30</td></tr><tr><td>Regrowable</td><td>No</td></tr><tr><td>Days to Re-Grow</td><td>0</td></tr></table>	Seed Price	50	Selling Price	50	Days to Grow	30	Regrowable	No	Days to Re-Grow	0
Seed Price	50											
Selling Price	50											
Days to Grow	30											
Regrowable	No											
Days to Re-Grow	0											
Mushrooms 	Mushroom Seed 	<table><tr><td>Seed Price</td><td>50</td></tr><tr><td>Selling Price</td><td>60</td></tr><tr><td>Days to Grow</td><td>45</td></tr><tr><td>Regrowable</td><td>No</td></tr><tr><td>Days to Re-Grow</td><td>0</td></tr></table>	Seed Price	50	Selling Price	60	Days to Grow	45	Regrowable	No	Days to Re-Grow	0
Seed Price	50											
Selling Price	60											
Days to Grow	45											
Regrowable	No											
Days to Re-Grow	0											
Onion 	Onion Seed 	<table><tr><td>Seed Price</td><td>30</td></tr><tr><td>Selling Price</td><td>25</td></tr><tr><td>Days to Grow</td><td>45</td></tr><tr><td>Regrowable</td><td>No</td></tr><tr><td>Days to Re-Grow</td><td>0</td></tr></table>	Seed Price	30	Selling Price	25	Days to Grow	45	Regrowable	No	Days to Re-Grow	0
Seed Price	30											
Selling Price	25											
Days to Grow	45											
Regrowable	No											
Days to Re-Grow	0											
Potato	Potato Seed	<table><tr><td>Seed Price</td><td>50</td></tr><tr><td>Selling Price</td><td>75</td></tr><tr><td>Days to Grow</td><td>60</td></tr></table>	Seed Price	50	Selling Price	75	Days to Grow	60				
Seed Price	50											
Selling Price	75											
Days to Grow	60											

		<table><tr><td>Regrowable</td><td>No</td></tr><tr><td>Days to Re-Grow</td><td>0</td></tr></table>	Regrowable	No	Days to Re-Grow	0						
Regrowable	No											
Days to Re-Grow	0											
Scallion 	Scallion Seed 	<table><tr><td>Seed Price</td><td>25</td></tr><tr><td>Selling Price</td><td>45</td></tr><tr><td>Days to Grow</td><td>30</td></tr><tr><td>Regrowable</td><td>No</td></tr><tr><td>Days to Re-Grow</td><td>0</td></tr></table>	Seed Price	25	Selling Price	45	Days to Grow	30	Regrowable	No	Days to Re-Grow	0
Seed Price	25											
Selling Price	45											
Days to Grow	30											
Regrowable	No											
Days to Re-Grow	0											
Strawberry 	Strawberry Seed 	<table><tr><td>Seed Price</td><td>75</td></tr><tr><td>Selling Price</td><td>95</td></tr><tr><td>Days to Grow</td><td>30</td></tr><tr><td>Regrowable</td><td>Yes</td></tr><tr><td>Days to Re-Grow</td><td>7</td></tr></table>	Seed Price	75	Selling Price	95	Days to Grow	30	Regrowable	Yes	Days to Re-Grow	7
Seed Price	75											
Selling Price	95											
Days to Grow	30											
Regrowable	Yes											
Days to Re-Grow	7											
Bell Pepper 	Bell Pepper 	<table><tr><td>Seed Price</td><td>50</td></tr><tr><td>Selling Price</td><td>75</td></tr><tr><td>Days to Grow</td><td>60</td></tr><tr><td>Regrowable</td><td>Yes</td></tr><tr><td>Days to</td><td>10</td></tr></table>	Seed Price	50	Selling Price	75	Days to Grow	60	Regrowable	Yes	Days to	10
Seed Price	50											
Selling Price	75											
Days to Grow	60											
Regrowable	Yes											
Days to	10											

		<table><tr><td>Re-Grow</td><td></td></tr></table>	Re-Grow									
Re-Grow												
Tomato 	Tomato Seed 	<table><tr><td>Seed Price</td><td>80</td></tr><tr><td>Selling Price</td><td>65</td></tr><tr><td>Days to Grow</td><td>45</td></tr><tr><td>Regrowable</td><td>Yes</td></tr><tr><td>Days to Re-Grow</td><td>7</td></tr></table>	Seed Price	80	Selling Price	65	Days to Grow	45	Regrowable	Yes	Days to Re-Grow	7
Seed Price	80											
Selling Price	65											
Days to Grow	45											
Regrowable	Yes											
Days to Re-Grow	7											

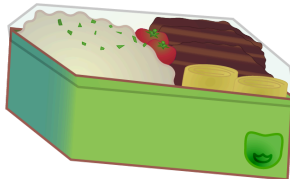
LIST OF PLANTING EQUIPMENTS

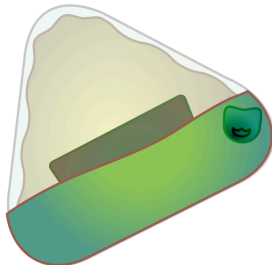
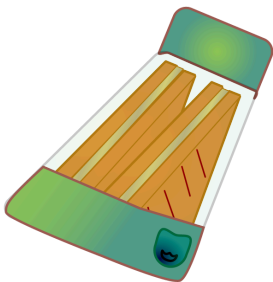
Tools	Description	Attributes		
Gloves 	Used to harvest the Fruits and Vegetables.	<table><tr><td>Stamina Cost</td><td>5</td></tr></table>	Stamina Cost	5
Stamina Cost	5			
Hoe 	Use to remove Plants or Dead Plants.	<table><tr><td>Stamina Cost</td><td>5</td></tr></table>	Stamina Cost	5
Stamina Cost	5			

Shovel 	It is used to Dig up the soil in order to plant and can also be used on removing weeds.	Stamina Cost 5
Watering Can 	Waters the Soil for the Plant to Grow	Stamina Cost 5

Consumables

LIST OF FOODS THAT CAN BE BOUGHT IN THE CONVENIENCE STORE

Foods and Drinks	Description	Attributes						
BENTO 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>100</td></tr><tr><td>Energy Refill</td><td>15</td></tr><tr><td>Hunger Refill</td><td>50</td></tr></table>	Buying Price	100	Energy Refill	15	Hunger Refill	50
Buying Price	100							
Energy Refill	15							
Hunger Refill	50							
CHOCO 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>20</td></tr><tr><td>Energy Refill</td><td>15</td></tr><tr><td>Hunger Refill</td><td>5</td></tr></table>	Buying Price	20	Energy Refill	15	Hunger Refill	5
Buying Price	20							
Energy Refill	15							
Hunger Refill	5							

RICE BALL 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>50</td></tr><tr><td>Energy Refill</td><td>15</td></tr><tr><td>Hunger Refill</td><td>25</td></tr></table>	Buying Price	50	Energy Refill	15	Hunger Refill	25
Buying Price	50							
Energy Refill	15							
Hunger Refill	25							
SANDWICH 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>35</td></tr><tr><td>Energy Refill</td><td>20</td></tr><tr><td>Hunger Refill</td><td>10</td></tr></table>	Buying Price	35	Energy Refill	20	Hunger Refill	10
Buying Price	35							
Energy Refill	20							
Hunger Refill	10							
SODA CAN 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>50</td></tr><tr><td>Energy Refill</td><td>20</td></tr><tr><td>Hunger Refill</td><td>-5</td></tr></table>	Buying Price	50	Energy Refill	20	Hunger Refill	-5
Buying Price	50							
Energy Refill	20							
Hunger Refill	-5							
CANNED MEAT 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>75</td></tr><tr><td>Energy Refill</td><td>15</td></tr><tr><td>Hunger Refill</td><td>25</td></tr></table>	Buying Price	75	Energy Refill	15	Hunger Refill	25
Buying Price	75							
Energy Refill	15							
Hunger Refill	25							

CHIPS 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>15</td></tr><tr><td>Energy Refill</td><td>5</td></tr><tr><td>Hunger Refill</td><td>15</td></tr></table>	Buying Price	15	Energy Refill	5	Hunger Refill	15
Buying Price	15							
Energy Refill	5							
Hunger Refill	15							
WATER 	Purchased in the Convenience Store	<table><tr><td>Buying Price</td><td>20</td></tr><tr><td>Energy Refill</td><td>45</td></tr><tr><td>Hunger Refill</td><td>5</td></tr></table>	Buying Price	20	Energy Refill	45	Hunger Refill	5
Buying Price	20							
Energy Refill	45							
Hunger Refill	5							

LIST OF FOODS THAT CAN BE BOUGHT FROM THE CHEF

Food	Description	Attributes						
Salad 	Cooked by the Chef List of Ingredients - Lettuce x2 - Tomato x3	<table><tr><td>Buying Price</td><td>75</td></tr><tr><td>Energy Refill</td><td>25</td></tr><tr><td>Hunger Refill</td><td>30</td></tr></table>	Buying Price	75	Energy Refill	25	Hunger Refill	30
Buying Price	75							
Energy Refill	25							
Hunger Refill	30							

Veggie Pops 	Cooked by the Chef List of Ingredients - Lettuce x3 - Tomato x3	<table><tr><td>Buying Price</td><td>75</td></tr><tr><td>Energy Refill</td><td>20</td></tr><tr><td>Hunger Refill</td><td>50</td></tr></table>	Buying Price	75	Energy Refill	20	Hunger Refill	50
Buying Price	75							
Energy Refill	20							
Hunger Refill	50							
Fried Mushroom	Cooked by the Chef List of Ingredients Mushroom x5	<table><tr><td>Buying Price</td><td>75</td></tr><tr><td>Energy Refill</td><td>25</td></tr><tr><td>Hunger Refill</td><td>75</td></tr></table>	Buying Price	75	Energy Refill	25	Hunger Refill	75
Buying Price	75							
Energy Refill	25							
Hunger Refill	75							
Vegetable Fried Rice	Cooked by the Chef List of Ingredients - Scallion x2 - Lettuce x1 - Mushroom x1	<table><tr><td>Buying Price</td><td>30</td></tr><tr><td>Energy Refill</td><td>50</td></tr><tr><td>Hunger Refill</td><td>60</td></tr></table>	Buying Price	30	Energy Refill	50	Hunger Refill	60
Buying Price	30							
Energy Refill	50							
Hunger Refill	60							

LIST OF FOODS THAT CAN BE COOKED BY THE PLAYER

Food	Description	Attributes	
Soup	Cooked by Player List of Ingredients - Lettuce x3 - Mushroom x2 - Potato x1		
		Energy Refill	20
		Hunger Refill	40

Salad	Cooked by Player List of Ingredients • Lettuce x3 • Garlic x2 • Scallion x1 • Onion x1	<table><tr><td>Energy Refill</td><td>25</td></tr><tr><td>Hunger Refill</td><td>35</td></tr></table>	Energy Refill	25	Hunger Refill	35
Energy Refill	25					
Hunger Refill	35					
Stir Fried Veggies	Cooked by Player List of Ingredients • Garlic x1 • Lettuce x4 • Tomato x2	<table><tr><td>Energy Refill</td><td>25</td></tr><tr><td>Hunger Refill</td><td>30</td></tr></table>	Energy Refill	25	Hunger Refill	30
Energy Refill	25					
Hunger Refill	30					
Tomato Soup	Cooked by Player List of Ingredients • Tomato x3 • Scallion x2	<table><tr><td>Energy Refill</td><td>35</td></tr><tr><td>Hunger Refill</td><td>45</td></tr></table>	Energy Refill	35	Hunger Refill	45
Energy Refill	35					
Hunger Refill	45					

2.3 Secondary Mechanics

Company Minigame & Payday

Company Minigame: The player can earn extra money by completing the minigame, regardless of winning or losing. The amount earned is based on the number of pillars the player successfully passes through.

--The minigame can only be played once per day.

--Upon completion, the in-game time will automatically skip to 5:00 PM.

Payday: The player also receives a fixed income from their job on Saturday, which is automatically added to their money.

Earning Money: The player can earn money through the following methods:

1. Company Minigame – Participate in the Company Minigame to earn extra money.
2. Payday – Passively receive money every week.
3. Selling Harvested Crops – Sell crops that have been harvested to generate income.

Reduce Money: The player's money decreases through various expenses, including:

1. Purchases – Buying items, crops, or other goods.
2. Medical Bills – A fixed cost of 500 for medical expenses when the Energy of Hunger reaches zero
3. Apartment Rent – A variable rent cost ranging from 500 to 1000 on every 30th day of the month

2.4 Additional features

Kitty Kultivation simulates real life gardening to an extent as it shows which plants can be planted and how they are taken care of. Although some plants require different approaches to how they are prepared and kept healthy, Kitty Kultivation will NOT properly and accurately represent the proper care management for each individual plant. There will be individual growth time that would be lessened for game purpose and each plant will have its own unique look. Kitty Kultivation will however, contain a codex where the players can access through the journal, where both in-game instruction and real life instructions on how the plants are taken care of is located, there will also be scientific descriptions of what the plant is, its scientific name, and a fun fact about the plant.

2.5 Resources

The Following are the resources the player will have to manage throughout the game:

- **Energy**

- The energy dictates whether or not the player has the ability to do an action each day. Energy will start each day at max (100) and actions such as Planting, Watering, Removing Plant and Weeds, Harvesting will decrease the current amount of energy the player has. There will be items in the game that the player can consume to recover a certain amount of energy.
- When the player reaches 0 energy they will be warned that doing any more actions will cause their character to pass out and it is recommended that the player should go to Bed rather than doing anymore work.
- Every gardening action will cost **5 energy**.
- REPLENISH ENERGY
 - Navigate to **Consumables** (for Energy and Hunger refill)
 - Sleeping = 100

- **Hunger**

- The Hunger Meter is a secondary resource of the game, however if it is ignored by the player there will be consequences. Food can be used to recover the Hunger, and as long as the player has crops or money to buy food, they can manage the Hunger. Hunger Meter decreases overtime.
- When the Hunger Meter falls to zero, the player will begin losing energy at a higher rate than normal.
- REPLENISH HUNGER
 - Navigate to **Consumables** (for Energy and Hunger refill)

- **Money**

- Money is an important resource in the game since it can dictate whether the player loses the game or not. Money can be earned either through working and receiving money after a week, or selling harvested crops.
- Money is used to pay rent on Day 30. If the player can't pay, they're evicted, resulting in Game Over.
- Money can also be used to purchase. SEEDS, GARDEN UPGRADES, FOODS, FRUITS and VEGETABLE.

2.6 Economy

The crops harvested can be resold by the player at the same price they were bought at, hence ensuring a good turnover. Seeds, however, are reasonably priced, and it is worth the player's money to buy and plant them since the yield after harvesting will be a good business.

Each Saturday, the player earns a random passive income as a reward, symbolizing other income from miscellaneous sources. Otherwise, the player can play a minigame (Work) to earn additional money from active play.

In order to extend their farming capability, the player can invest in an increase in the number of planting plots. With the expansions, there will be the capacity to plant more crops at the same time, which translates to more money. Upgrading is accomplished by buying expansions as in-game money or by achieving farming milestones to open up additional plots of land.

2.7 Losing

Kitty Kultivation was originally planned to be similar to stardew valley, where the player accomplishes tasks without the risk of losing everything. We thought about a similar approach to this mechanic, The player has a hunger bar and energy bar in which they must maintain or else will be charged a hefty amount of money and will be teleported to the Bedroom. When the player has run out of money and the

day of rent has come, the player will be evicted from the apartment resulting in Game Over.

2.8 Setting {Imagine Animal Crossing but only the gardening and inside a city}

Kitty Kultivation is set in a world where the downsides of urbanization are visible and is still partially unaddressed. The scene is similar to downtown New York with pollution similar to that of cities in India, Bangladesh or Pakistan. The player will have their own apartment that would be a contrast to the atmosphere outside.

2.9 Story

In the smog and polluted ridden city of Kat Town, a weary office cat spends their days trapped in the soul crushing corporate life. Until one day, our cat finds an old journal from their countryside grandparent that changes everything. Filled with sketches of the farm fields and handwritten notes about the quiet joy of growing things, it sparks an idea for our office cat. But leaving the city isn't an option due to their work, so they make a radical decision : “If I can't leave the city and go to the farm, I'll bring the farm to me”. Armed with a packet of leftover seeds, some tools bought in the supermarket and pots from fake plants, our city cat transforms their cramped apartment into a green oasis, defying the mundane and concrete world outside.

3 Art

3.1 Theme Interpretation

The game maintains a pastel color theme or vibrant color theme when inside the apartment but when the player is outside the colors would resemble a polluted city, darker or grayed colors. A great inspiration for the game when outside is EliasDaler's Project M.T.P.



3.2 Design

The design is mostly low poly with a stylized design that seems a bit more modern. Main inspiration for how the design are made are astroneer or EliasDaler's Project M.T.P

4 Audio

4.1 Music

To add to the theme of the game, Kitty Kultivation will take direct inspiration from outer wild's music direction, especially the acoustics. But the music is mainly Bossa Nova.



4.2 Sound Effect

Considering that the game takes place in a bustling city, ambient music would be played alongside environmental sound effects, when in an alley, the ambient music would be lower and there would be changes in sound effects like dripping water or the exhaust vents.

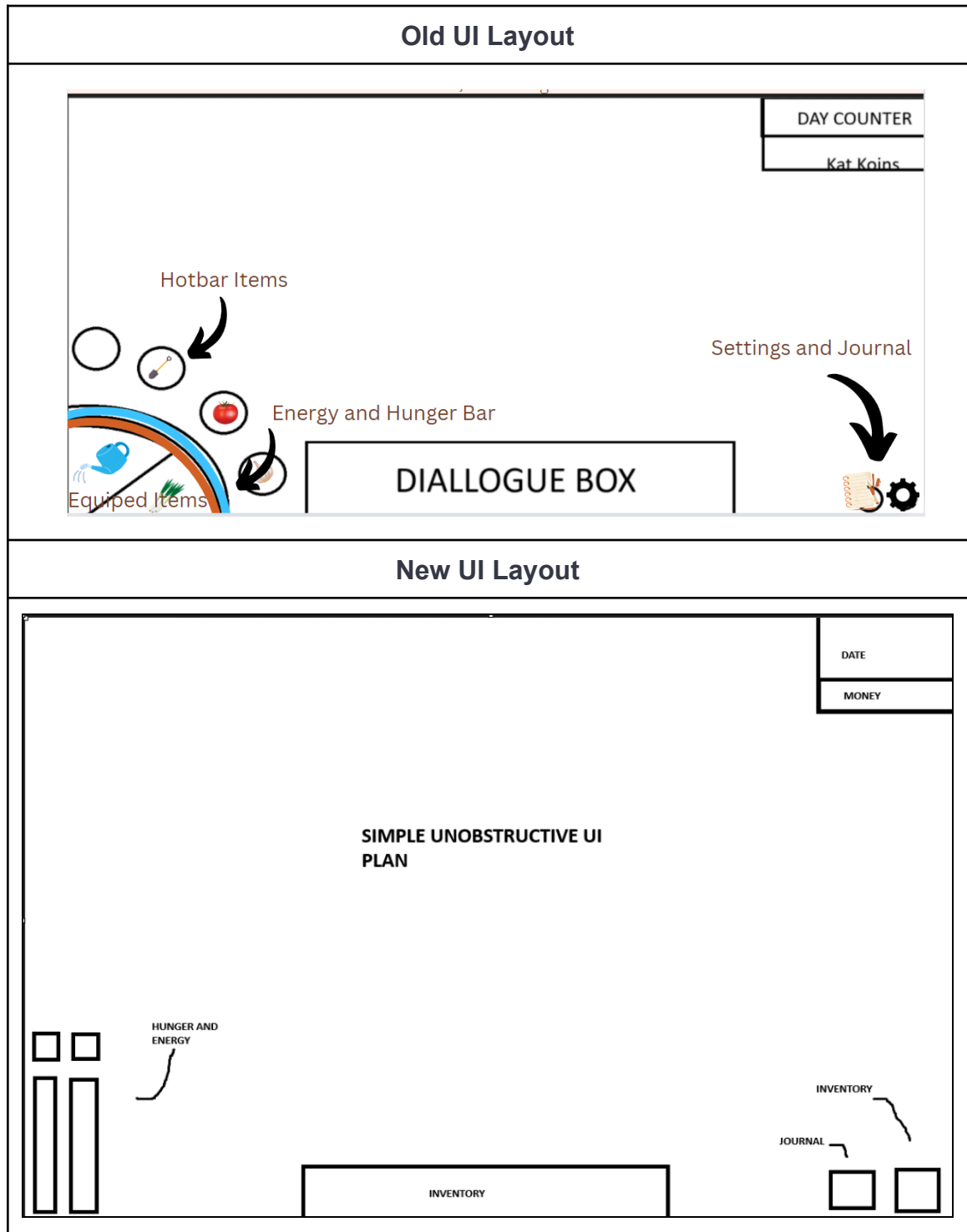


5 Game Experience

5.1 UI

The UI would incorporate the same artstyle of the game, with more bubbly designs, emphasizing the theme of the game





5.2 Controls

Keyboard

Arrow keys / WASD - Movement

F - Interact

I - Inventory

O - Journal

6 Concept Art and References



A Game Created by user EliasDaler in Twitter/X

[A heavy inspiration for the art direction of the Kitty Kultivation]

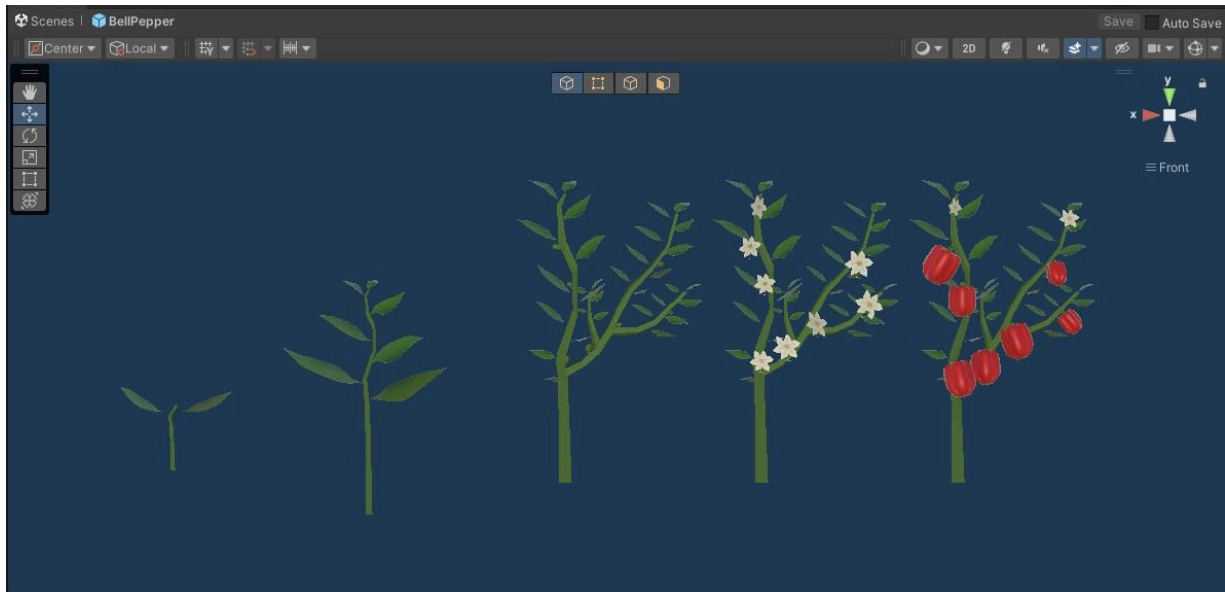


Neco Arc's Character Design is a major inspiration for our character design, which resembles the character in some parts.

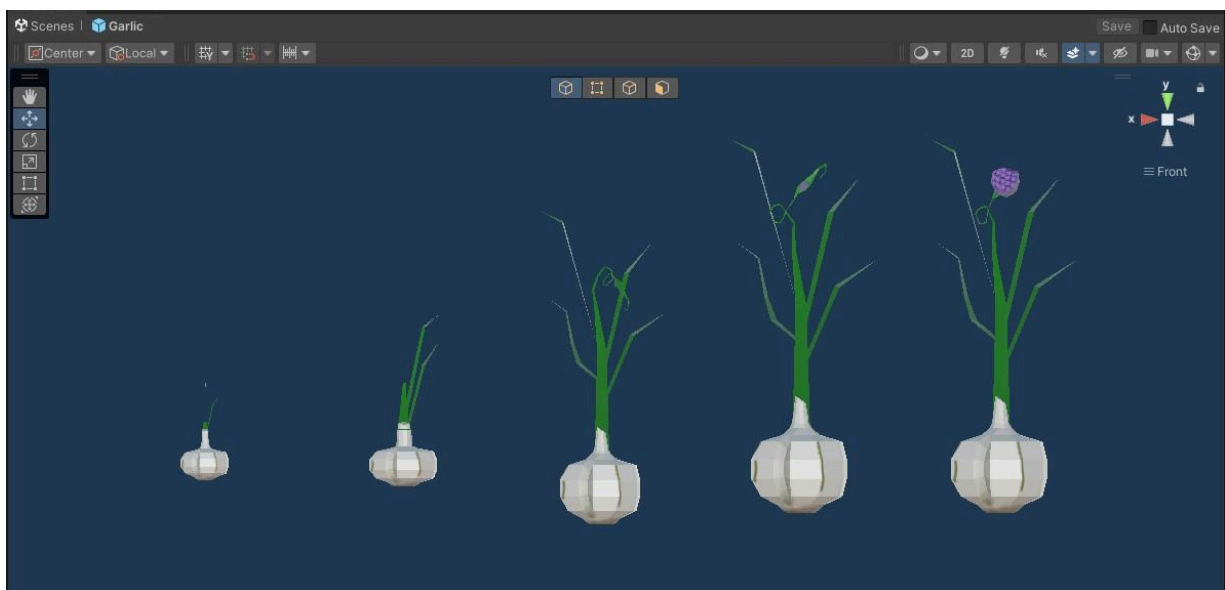
7 Game Output

7.1 Plant Models

Bell Pepper



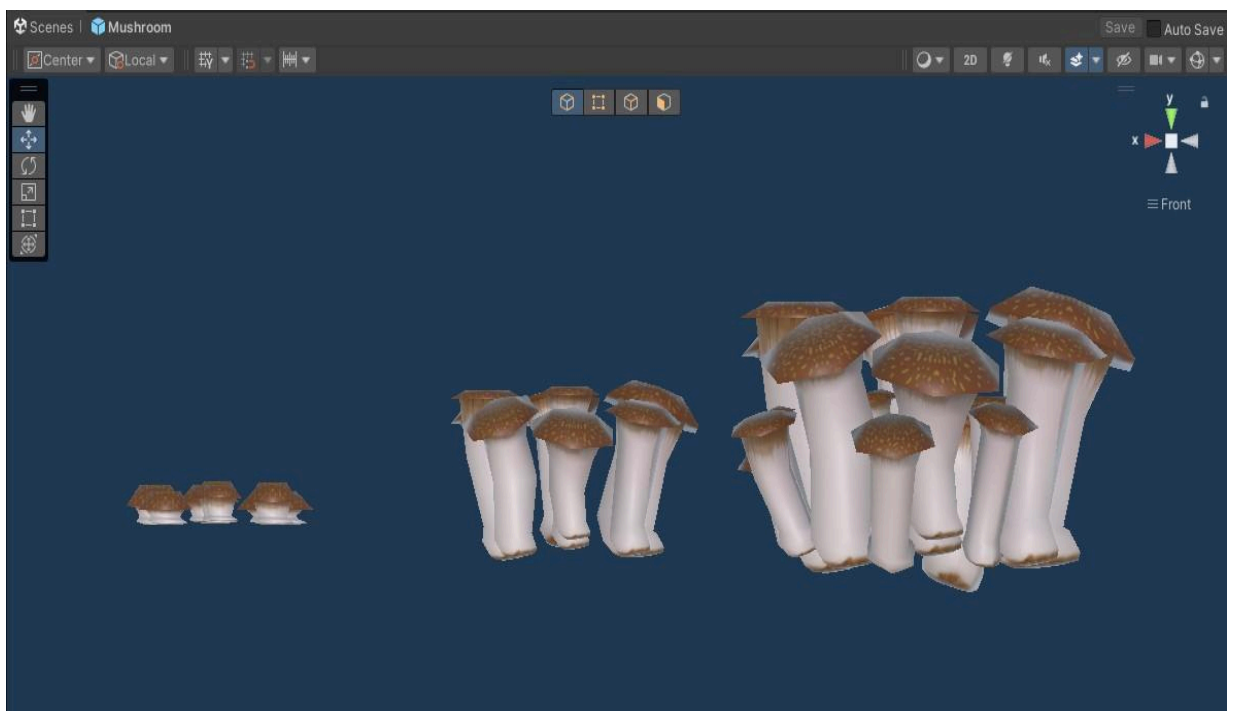
Garlic



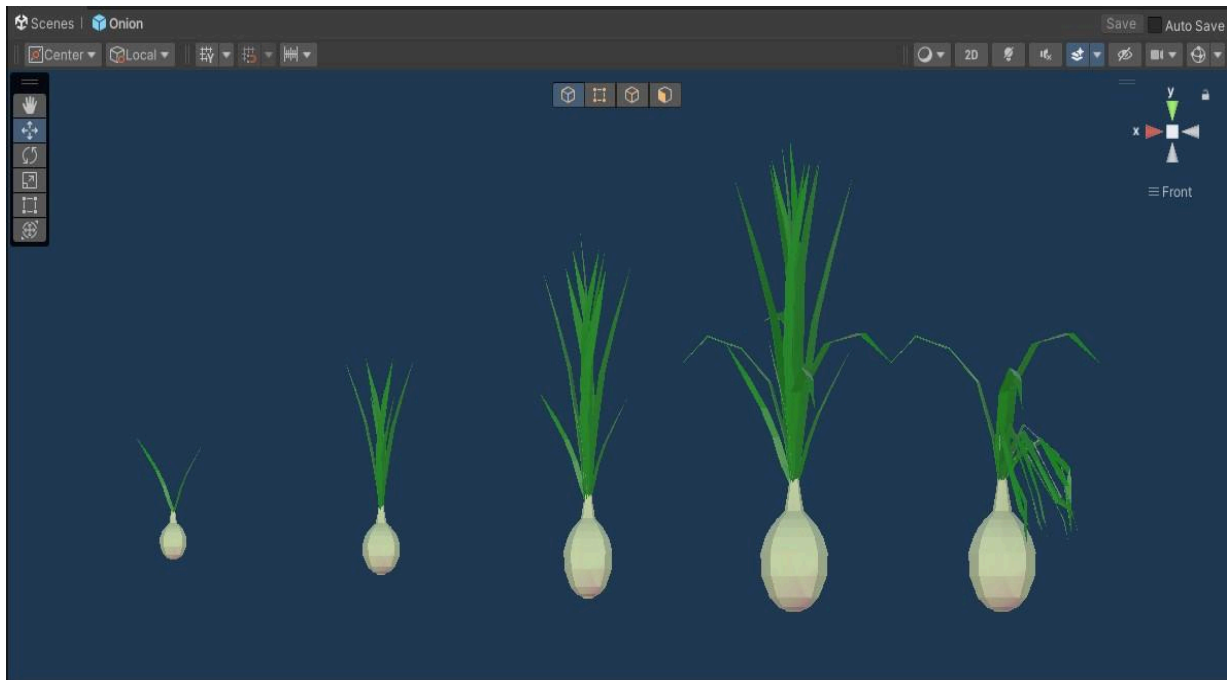
Lemon



Mushroom

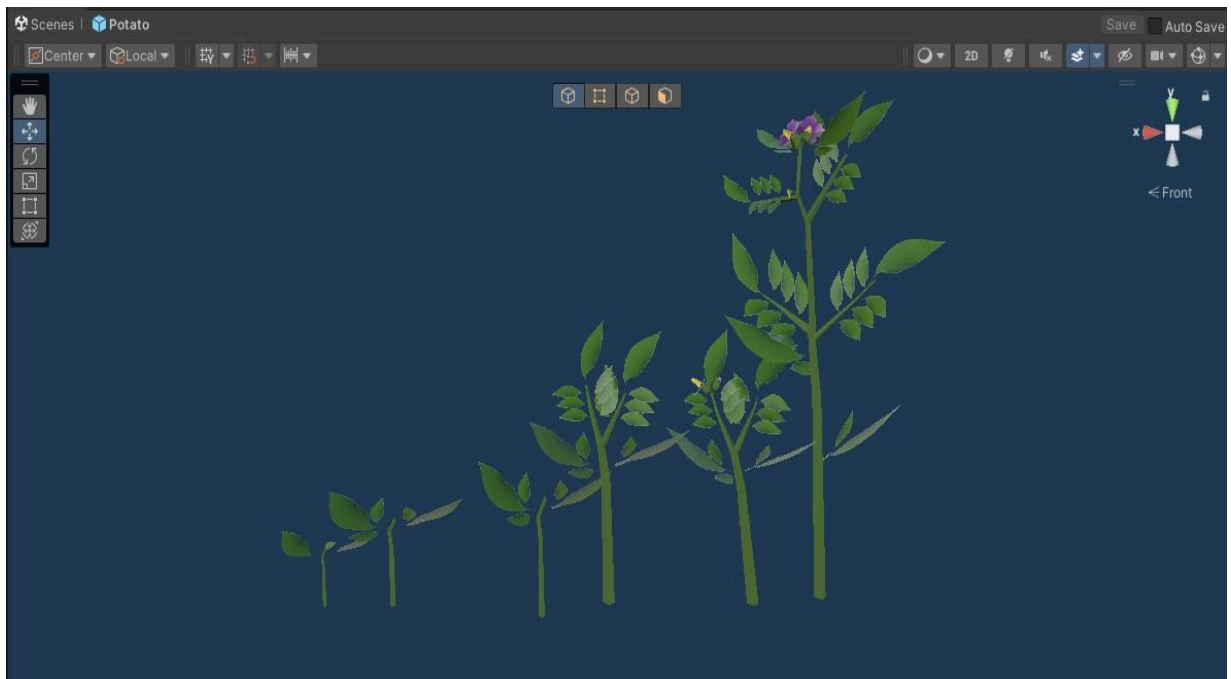


Onion

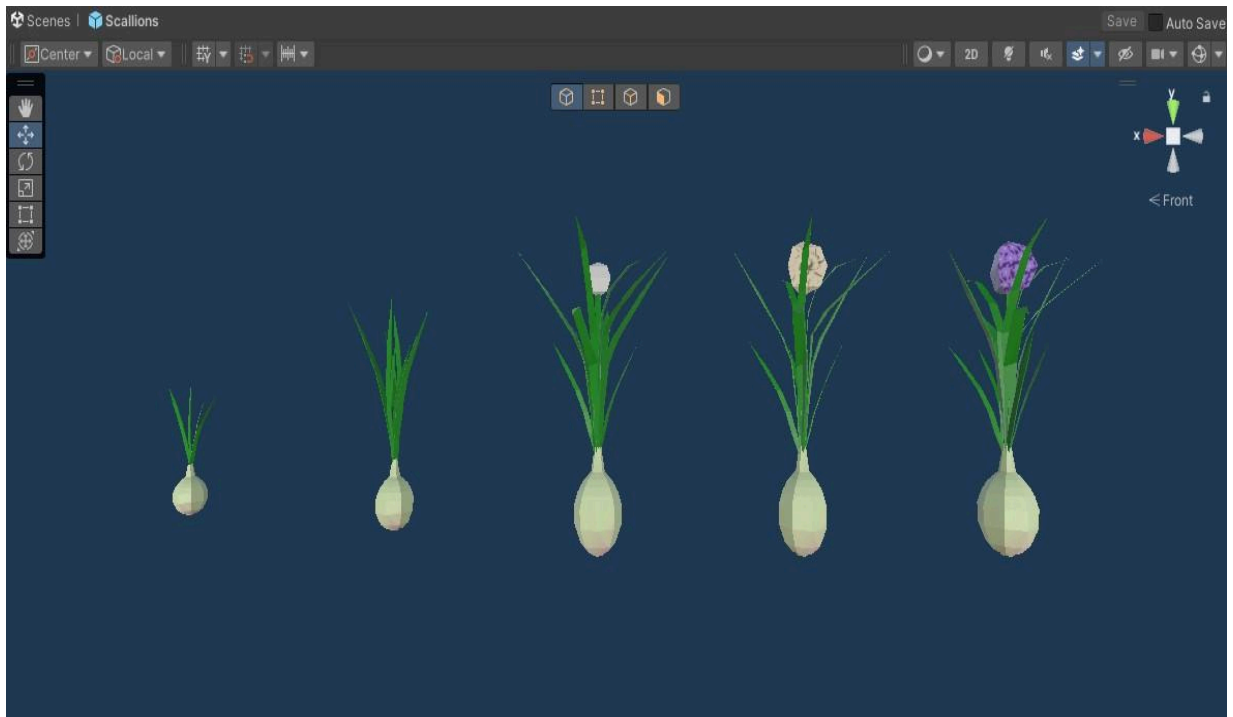


Potato

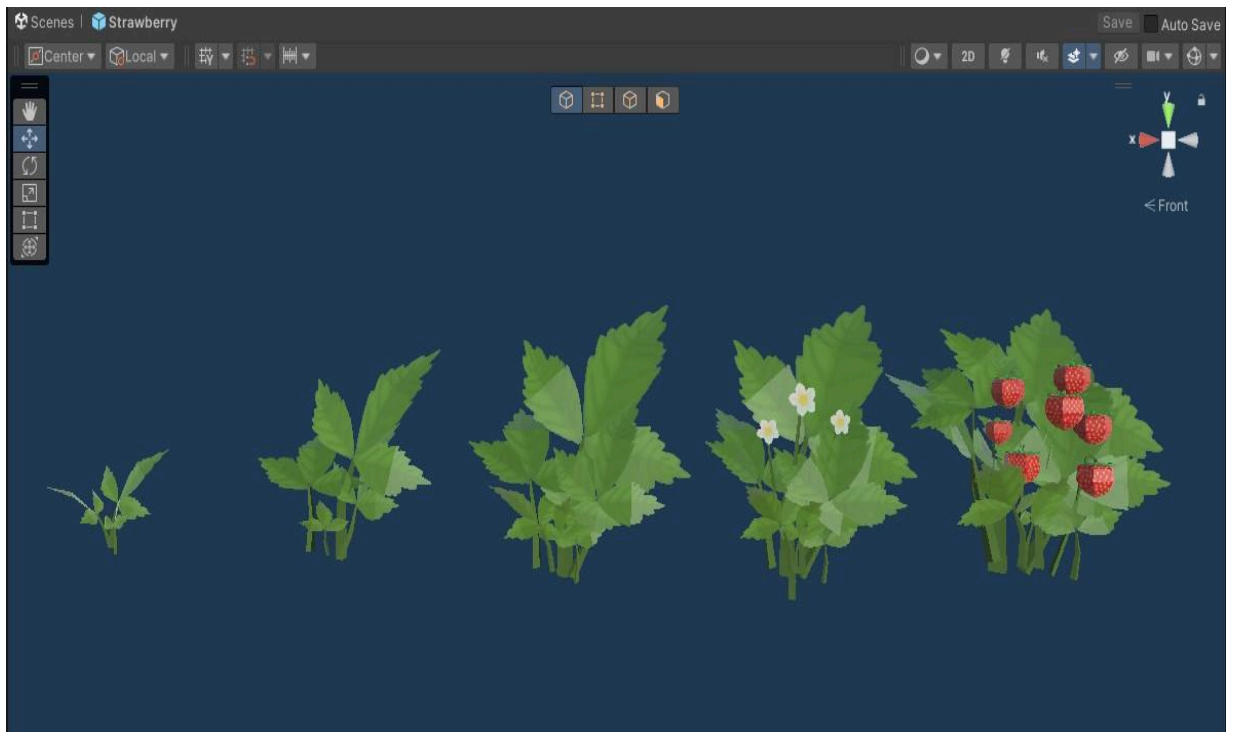
(Since the potato yield is normally below the ground, we decided to not include it in the model)



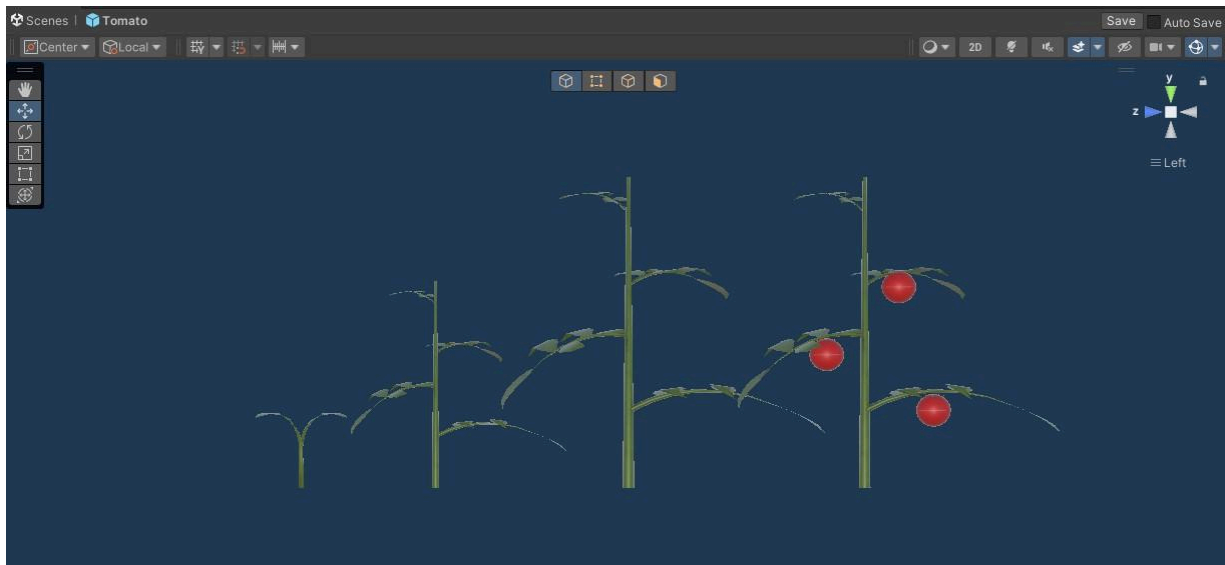
Scallion



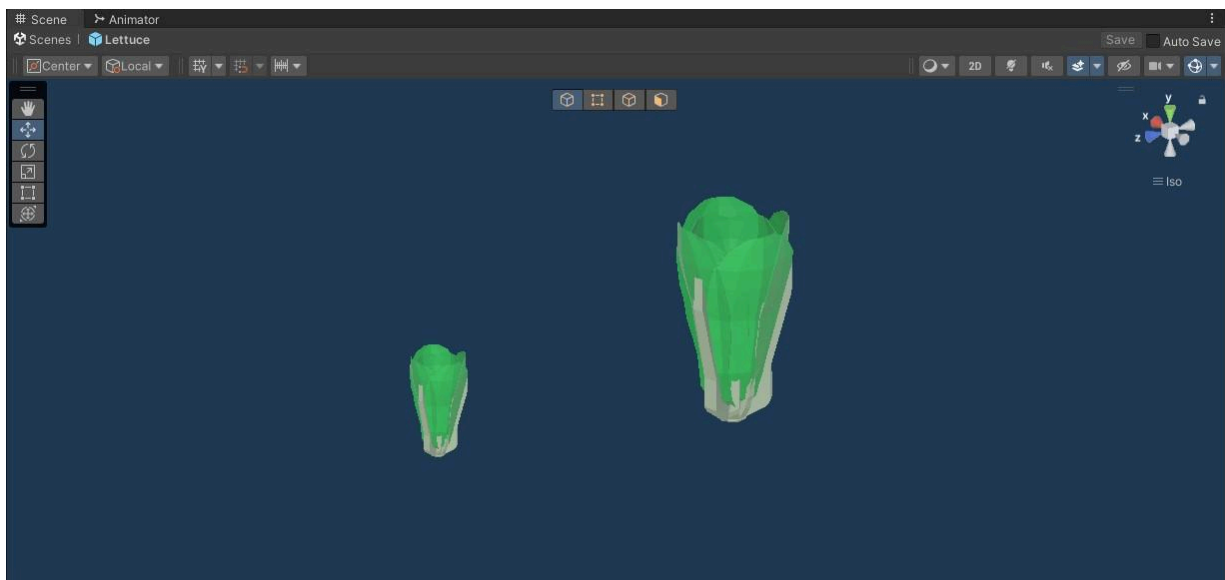
Strawberry



Tomato



Lettuce



7.2 Character and NPC's

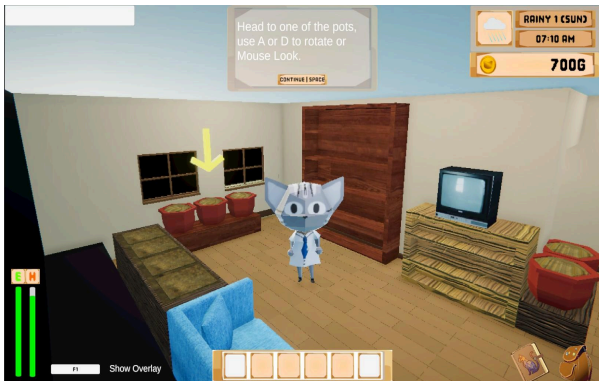
Player	Upgrade Store Clerk
	
Convenience Store Clerk	Chef
	
Greengrocer (Seller of Fruits/Vegetables and Seeds)	Buyer (Fruits and Vegetables Buyer)

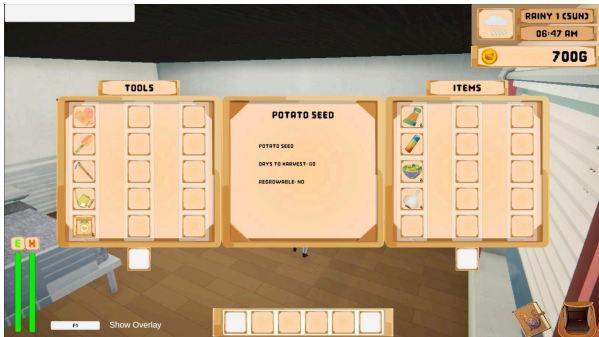
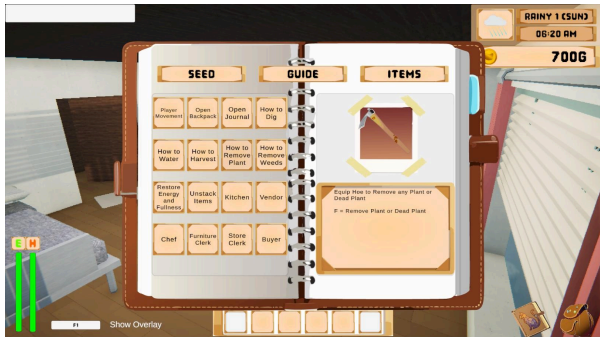
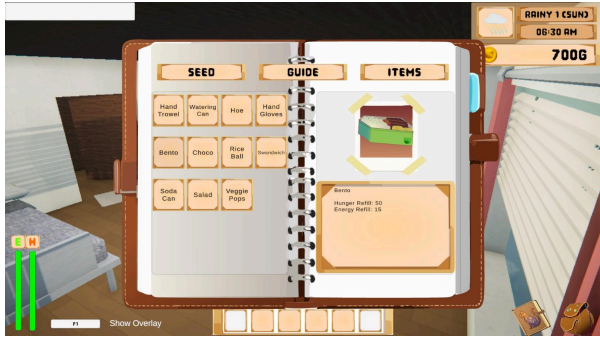


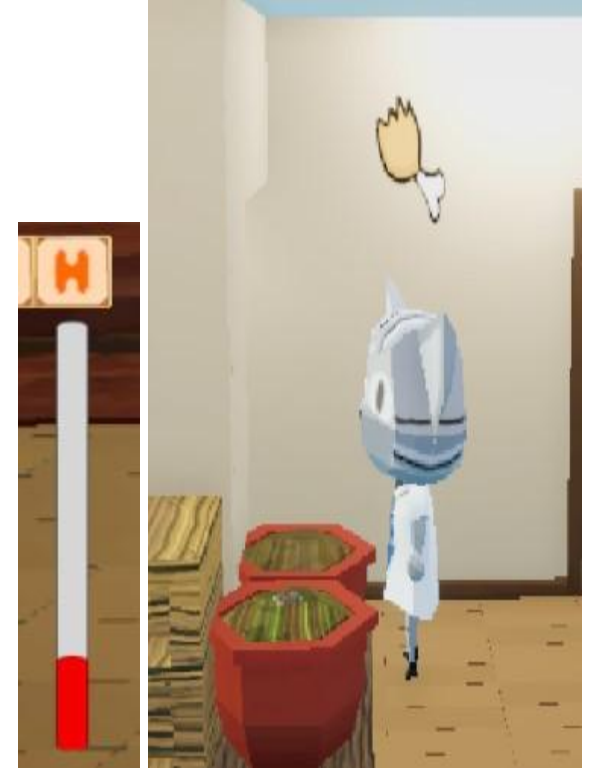
Random NPC's

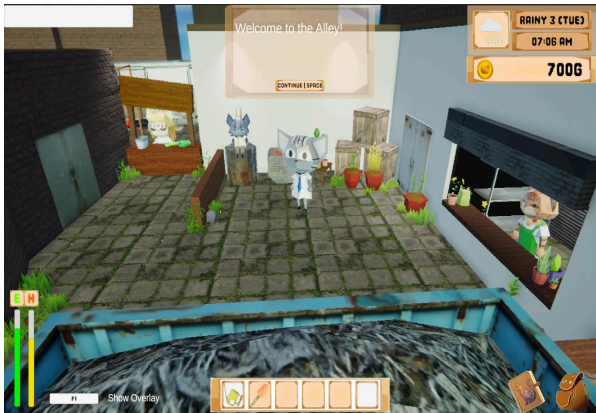


7.3 Game Screenshots

Main Menu 	Bedroom 
Tutorial Section (Instructions Middle Top) 	City Guide 
Planting Area 	Inventory 
Inventory when Item is Hovered	Journal Guide

	
Journal Seed	Journal Items
	
Plant Information when Planted	Need Water Notification
	
Remove Weed Notification	Dead Plant

	
Depleted Hunger (Blinking)	Depleted Energy (Blinking)
	
Backalley	Buyer of Harvest Fruits and Vegetables



Selling Prompt



Dialogue



Fruit and Seed Vendor



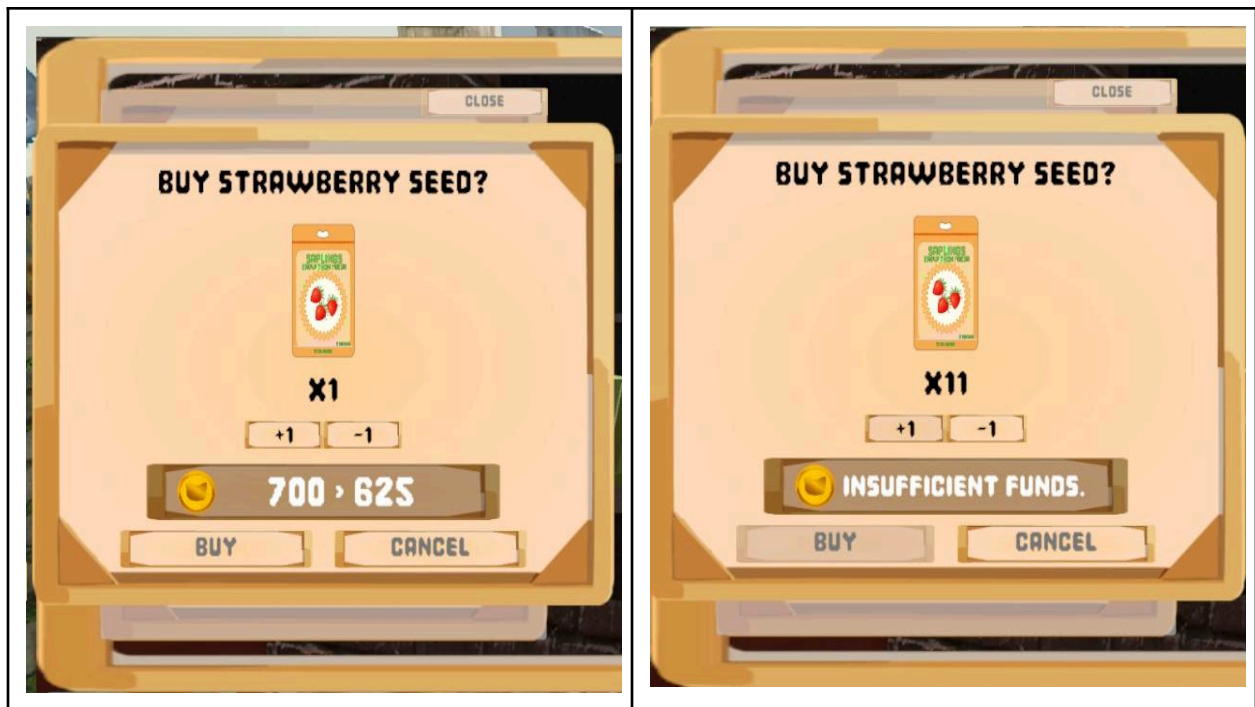
Selected an Item



Buying



Not Enough Money



Chef



Convenience Store



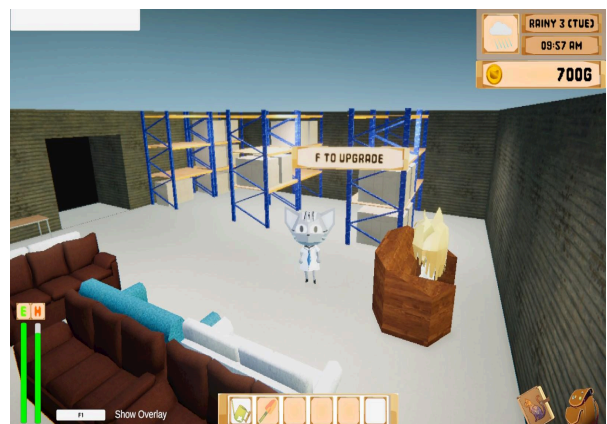
Store



Store Clerk



Upgrade Shop



Bus Stop (To Work)



Office



Work Place	Minigame
Minigame End	Kitchen

Story Panels

